

TUTORIAL 6

Introduction To Programming

```
file_present = open("intro_toProg.txt", "r")  
  
#print(file_present.read())  
  
print(file_present.readline())
```

Output

```
Welcome to the Introduction to Programming course!!
```

How can loops be used to read the line by line contents of a file?? Let's see!

```
file = open('intro_toProg.txt', 'r')  
print("Using for loop:")  
print("")  
for line in file:  
    print(line)
```

Output

```
Using for loop:  
  
Welcome to the Introduction to Programming course!!  
  
The stage is all yours, all you need is a desire to learn.
```

Similarly, TAs can show that to read 'n' number of characters, they can use read(n) function. (all the codes and text files will be shared to the TAs).

Using close()

TAs will explain why it is important to close files after the use. It eradicates the possibility, of unrequired updation and modifications that could cause errors.

```

print("Reading File after opening:")

file = open("intro_toProg.txt", "r")

print("Printing the contents")
print("")

print(file.read())

print("Closing The file")

file.close()

print("Trying to read the file again after closing!!")
print("It must show ERROR!!!")

print(file.read())

```

Output

```

Reading File after opening:
Printing the contents

Welcome to the Introduction to Programming course!!
The stage is all yours, all you need is a desire to learn.
Keep practicing!!
Closing The file
Trying to read the file again after closing!!
It must show ERROR!!!
Traceback (most recent call last):
  File "tutorial6.py", line 37, in <module>
    print(file.read())
ValueError: I/O operation on closed file.

```

This week's tutorial will start with the basics of file handling. First, the TAs would start with an error code which is trying to open a text file in read-mode, that doesn't exist:

```

## Let's read some files!!

## start with some error:->

file_notPresent = open("somefile.txt", 'r')
## 'r' represents read mode

```

Output

```

Traceback (most recent call last):
  File "tutorial6.py", line 5, in <module>
    file_notPresent = open("somefile.txt", 'r')
FileNotFoundError: [Errno 2] No such file or directory: 'somefile.txt'

```

But, there exists a file in the system, called 'intro_toProg.txt'. Let's open it and read its contents.

```

file_present = open("intro_toProg.txt", "r")

print(file_present.read())

```

Output

```
Welcome to the Introduction to Programming course!!  
The stage is all yours, all you need is a desire to learn.  
Keep practicing!!
```

TAs would then ask the students if, there was a way to read a single line?? **readline()**

The TAs would then proceed with follow-up questions, data for these questions is contained in text files that will be shared with them.

Q1. You are being provided a text file which contains the famous sonnet, OZYMANDIAS, written by the great P.B. Shelley.

You are required to:

- Find the number of lines in the text file.
- Find the longest word(s) in the poem.
- Find the frequency of each of the words in the poem and return it as a dictionary

Q2, Again, You have been provided with a file. But this time, due to some error, every character "i" has been mistakenly written as "j".

(This question will be provided as a task to the students, no solution as such would be provided for it)

- Replace those 'j's with 'i'.
- Create another file and copy the contents of the file(after correcting all 'j's to 'is'), but with a twist. This time, Everything should be in UPPERCASE, in the new file.